ASHUTOSH RANJAN | 23CS10004

B.Tech.(Hons.) in COMPUTER SCIENCE AND ENGINEERING

EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2027	B.TECH	IIT Kharagpur	9.34 / 10
2023	CBSE Class XII	Krishna Public School	92.4%
2021	CBSE Class X	Open Minds A Birla School	75%

PROJECTS

Automatic Timetable Management Application | Course Project | IIT Kharagpur **[Mar '25 - Apr '25]**

Under the guidance of Prof. Debasis Samanta and Prof. Jibesh Patra

Live link: https://atma-ar.vercel.app *GitHub Link: https://github.com/Ashutosh-Ranjan09/ATMA-Final*

- Developed **ATMA**, a full-stack **React/Next.js** application with **MongoDB** backend to efficiently manage courses, rooms, and departments
- Implemented **JWT-based authentication** using **Next.js middleware** for secure access, per-user data isolation, and session management
- Designed **RESTful APIs** for **CRUD** operations and seamlessly integrated a robust custom **automated timetable scheduling algorithm**
- Constructed a **grid-based UI** for adjustments, enabling users to override algorithm-generated timetables with constraint enforcement
- Enabled **one-click PDF export** of timetables using **html2canvas** and **jsPDF** for effortless sharing, distribution, and long-term archiving

CF-Connect | Self Project **[May '25 - Jun '25]**

Live link: https://cf-connect.vercel.app *GitHub Link: https://github.com/Ashutosh-Ranjan09/cf-connect*

- Developed **CF-Connect**, a full-stack **Next.js** and **MongoDB** application for **Codeforces** users to easily connect, track progress, and grow
- Leveraged the **Codeforces API** to reliably import user profiles, then visualize performance trends, and analyze problem-solving efficiency
- Built a secure **custom handle-linking mechanism** to verify Codeforces accounts without relying on third-party authentication services
- Introduced **problem recommendation** and **social features** like follow, profile discovery, and **practice insights** to support **rating growth**
- Designed a modern, responsive UI with **Tailwind CSS** and **Recharts**, supporting fast navigation, intuitive search and interactive analytics

Dijkstra Visualizer | Self Project **[Jun '25]**

Live link: https://dijkstra-visual.vercel.app *GitHub Link: https://github.com/Ashutosh-Ranjan09/Dijkstra-visualizer*

- Built an **interactive Dijkstra's algorithm visualizer** using **Next.js** to efficiently demonstrate realtime step-by-step shortest path traversal
- Enabled detailed **graph construction** with **node dragging**, **edge editing**, direction toggling, and robust real-time **weight validation** logic
- Visualized traversal through dynamic **animations**, **path highlighting**, unreachable-node handling, and accurate **final cost computation**
- Added intuitive controls for **speed adjustment**, **manual stepping**, on-screen instructions, and comprehensive exploration of **edge cases**

RSA Chat Application | Self Project **[Jun '24]**

GitHub Link: https://github.com/Ashutosh-Ranjan09/Terminal-Chat-Room

- Developed a chat system using **Python TCP sockets** and **Rivest–Shamir–Adleman encryption** for secure peer-to-peer communication
- Built a custom **RSA algorithm** with **key generation** and **character-level encryption/decryption** for secure end-to-end data transmission
- Used **threading** to enable simultaneous message sending and receiving, ensuring smooth and responsive bi-directional communication
- Utilized **pickle** for serializing encrypted data and keys to ensure smooth and secure transmission during custom **RSA-based** messaging

SKILLS AND EXPERTISE

- **Programming Languages : Proficient -** C | C++ **Intermediate -** Python | CSS | HTML | JavaScript **Familiar -** TypeScript | Bash | Gawk
- **Skills :** Data Structures and Algorithms | Object-Oriented Programming | Competitive Programming | System Programming
- **Frameworks and Libraries :** C++ STL | NumPy | Matplotlib | Pandas | Next.js | Express.js | Mongoose
- **Utilities :** VS Code | Jupyter Notebook | GIT/GitHub | Postman | MongoDB Compass | Microsoft Office

COURSEWORK INFORMATION

Computer Science : Programming and Data Structures* | Algorithms-I* | Software Engineering* | Systems Programming Laboratory | Switching Circuits and Logic Design* | Discrete Structures | Formal Language and Automata Theory
Mathematics : Probability and Statistics | Linear Algebra and Complex Analysis | Advanced Calculus | Mathematical Statistics
Others : Basic Electronics* | Signals and Systems | Electrical Technology | Engineering Lab *{*includes laboratory component}*

AWARDS AND ACHIEVEMENTS

- Secured an **All India Rank of 873** in **JEE Advanced 2023**, placing in the **top 0.5%** among **1,80,000+** eligible candidates across the nation
- Attained an **All India Rank of 809** in **JEE Main 2023**, ranking in the **top 0.07%** among **1,110,000+** registered participants across India
- Achieved a peak **Codeforces rating of 1495 (Specialist)** through consistent problem-solving and contest participation **(handle: AR009)**
- Reached a peak **CodeChef rating of 1733 (3-Star)** by competing in rated contests and improving problem-solving skills **(handle: ar009)**
- Ranked **1106** globally in **Codeforces Round 998 (Div. 3)**, placing in the **top 3%** among **38,000+** competitve programmers worldwide
- Solved **500+** problems on platforms like **LeetCode**, **Codeforces**, and **GeeksforGeeks**, demonstrating a diverse problem-solving skillset

EXTRA CURRICULAR ACTIVITIES

- Member of the **OpenSoft** team representing the **Meghnad Saha Hall of Residence** for the **General Championship Technology 2025**
- Participated in the **National Sports Organization Health and Fitness Program**; officially certified for **perfect attendance** from **Unit 3**
- Contributed actively to the **Inter-Hall Illumination** event as part of **Meghnad Saha Hall of Residence's** representation at **IIT Kharagpur**

!Self declared by the student, CDC could not verify the relevant documents